

Digital Mice: Inducing and Validating Controlled Cognitive Deficits in Frontier Language Models for Mechanism Design

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Abstract

Early-stage mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must run experiments on people before they can learn which mechanisms are robust to bounded rationality. We ask whether frontier reasoning models can serve as the economist’s analogue of the laboratory mouse—capable reasoners instructed into specific, controlled cognitive deficits whose behavior can be studied before human subjects are recruited. Using a canonical second-price sealed-bid auction (SPSB) and a $3 \times 3 \times 3$ panel of *digital mice* defined over three cognitive axes (contingent reasoning, forward planning, belief reasoning), we establish two facts. First, the deficits have *construct validity*: repairing each axis moves truthful bidding in the theory-predicted direction, monotonically and with the predicted sign in all five pre-registered directional tests. Second, the panel—augmented by a small set of targeted error strains—attains *calibration coverage*: it reproduces all five canonical SPSB human-mistake classes (truthful, moderate underbidding, overbidding, extreme overbidding, and near-zero bidding) at rates above their detection thresholds. These results support the use of instructed-deficit language models as a phase-I screening population for mechanism design.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity.

1 Introduction

Medicine studies humans, but it does not begin with them. Much early drug testing starts with mice, and some of those mice are deliberately engineered to lack a specific protein, receptor, or channel. The deficit is not a nuisance; it is the identification strategy. If a treatment succeeds or fails only when a particular channel is removed, the experiment isolates a mechanism that is entangled in an intact organism.

Economics has lacked this luxury. The behavior of interest in mechanism design is human cognition—contingent reasoning, forward planning, belief formation, and strategic sophistication—and these channels are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Natural and laboratory experiments rarely deliver a population missing exactly one cognitive module, so the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models change the experimental object. The strongest current reasoning models are, in simple games, often closer to a fully rational agent than an ordinary subject: they follow long chains of reasoning and compute through cases. That capability is precisely what makes them useful here. We start from a capable reasoning substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation, not through dependence on hidden quirks of one model. This yields *digital mice*: language-model agents with specific, engineered cognitive deficits.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality states that (i) produce the behavioral signatures predicted by cognitive theory and (ii) span the empirically documented space of human mistakes, in a canonical second-price sealed-bid auction?

Contribution. We answer both halves affirmatively for the SPSB environment. Over a $3 \times 3 \times 3$ panel of 27 strains (810 bidder-decisions), repairing each cognitive axis raises the truthful-bid rate by +0.23 (contingent reasoning), +0.14 (forward planning), and +0.05 (belief reasoning); all five pre-registered directional hypotheses pass their sign test. The panel, augmented by two targeted error strains, covers all five canonical SPSB human-mistake classes at rates above threshold. We are explicit about a boundary of the

Table 1: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; the lowest level is the engineered deficit.

Axis	Level 0 (deficit)	Level 1	Level 2 (repaired)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

construction: the clean procedural basis naturally produces truthful and underbidding behavior but not overbidding, which we recover with a small number of hand-specified heuristic strains. We treat the digital-mouse panel as phase-I screening evidence, not as a prediction of human population shares.

2 The Digital-Mouse Panel

Two layers. A digital mouse is defined by two layers. The *rational base* is a state-of-the-art thinking model used as the best available fully rational agent. *Defect induction* then imposes a specific bounded-rationality instruction on the model’s deliberation procedure. The construction is model-agnostic by design: the deficit is a reasoning-procedure control, so as base models improve the rational mouse improves and the induced deficits become cleaner.

The $3 \times 3 \times 3$ grid. The current cognitive basis varies three axes, each at three levels, giving 27 strains. Table 1 lists the axis definitions. The lowest level of each axis is the engineered deficit; the highest is the repaired state. The cell with all axes repaired is closest to the rational mouse; the all-axes-off cell is the most deficient.

3 Experimental Design

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. Under standard preferences truthful bidding is a weakly dominant strategy, which makes truthfulness a clean construct-validity target: a more capable mouse should bid closer to its value.

Strains and runs. We instantiate all 27 basis strains with the same base model (gpt-5.4-mini, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total). Because the clean procedural basis does not naturally generate *overbidding*, we additionally author a small set of targeted *error mice*—heuristic strains designed to express one documented mistake each (e.g. a “second-price overgeneralizer” that treats bidding above value as safe because the winner pays the runner-up bid, and a “payment panic” strain that bids near zero). Seven error strains are specified; two had completed runs at the time of this analysis and feed the calibration scorecard below.

Outcome metrics. For each bidder-decision we record the bid, the value, and the deviation $b - v$. We define a truthful bid as $|b - v| \leq 0.05$, moderate underbidding as $b < v - 0.05$ (excluding near-zero bids), overbidding as $b > v + 0.05$, extreme overbidding as $b - v \geq 5$, and near-zero bidding as $b \leq 0.1$ with $v > 5$. These thresholds operationalize the five canonical SPSB mistake classes used in the calibration step.

4 Result 1: Construct Validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then repairing an axis should move behavior toward the rational benchmark. Figure 1 plots the truthful-bid rate at each axis level, marginalizing over the other two axes. All three axes are monotone or near-monotone in the repairing direction, and the contingent-reasoning axis produces the largest swing.

Table 2 reports the five pre-registered directional tests. For each axis we test that repairing it raises the truthful rate (expected positive) and, where applicable, lowers the underbidding rate (expected negative). All five tests pass: the realized sign matches the prediction and exceeds the minimum-magnitude bar in every case. The directional structure is exactly what theory predicts—more contingent reasoning and more forward planning push the agent toward its dominant strategy—which is the core construct-validity claim.

Figure 2 shows that the strains are also *separable* at the level of raw behavior. The rational mouse (contingent reasoning engaged) bids on the 45° truthful line; the all-deficits-off mouse systematically underbids; and the targeted error mice occupy distinct regions—persistent overbidding above the line and near-zero bidding along the floor. The mean absolute deviation falls from 2.78 for the most deficient basis strain to essentially zero (0.003) for the most rational one.

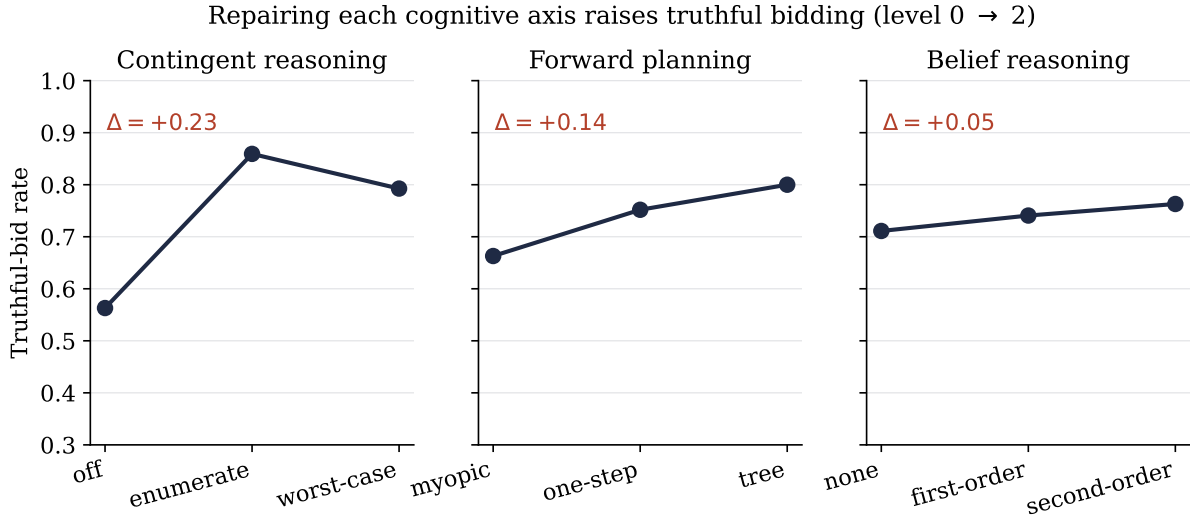


Figure 1: Construct-validity marginals. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Repairing each axis raises truthfulness, with the largest effect on the contingent-reasoning axis.

Table 2: Construct-validity diagnostics: five pre-registered directional tests on the cognitive axes. Δ is the level-2-minus-level-0 change in the metric. All tests pass.

Axis	Metric	Δ (L2–L0)	$ \Delta $ bar	Expected sign	Pass
Contingent reasoning	truthful rate	+0.230	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	0.05	negative	✓
Forward planning	truthful rate	+0.137	0.05	positive	✓
Forward planning	underbid rate	−0.137	0.05	negative	✓
Belief reasoning	truthful rate	+0.052	0.03	positive	✓

Notes. Sign rate = 1.00 (5 of 5). Underlying level means (truthful rate): contingent 0.56 $\rightarrow$ 0.86 $\rightarrow$ 0.79; forward 0.66 $\rightarrow$ 0.75 $\rightarrow$ 0.80; beliefs 0.71 $\rightarrow$ 0.74 $\rightarrow$ 0.76.

5 Result 2: Calibration Coverage

Construct validity establishes that the axes do what theory says. Calibration asks a different question: does the panel *span* the space of mistakes that real people make in this mechanism? We score five canonical SPSB human-mistake classes and ask, for each, whether some strain in the panel reproduces it at a rate above a detection threshold.

Figure 3 and Table 3 report the scorecard. All five classes are covered. Truthful and overbidding behavior are reproduced at very high rates (0.97 and 1.00); moderate underbidding, extreme overbidding, and near-zero bidding are reproduced at plausible intermediate rates (0.63, 0.44, 0.78). The best strain for each class is reported in the table: truthfulness and underbidding are covered by the clean procedural basis, whereas

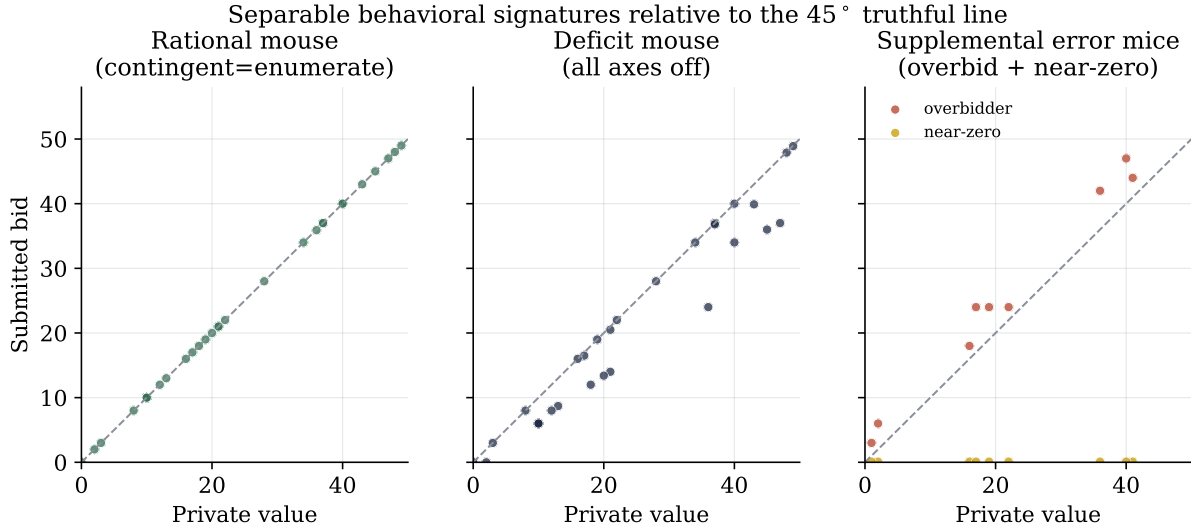


Figure 2: Separable behavioral signatures. Bid versus private value for a rational basis strain, the all-deficits-off strain, and the two supplemental error mice. The dashed line is the 45° truthful benchmark.

the overbidding and near-zero classes are covered by the targeted error mice—making explicit which classes the procedural construction reaches on its own and which require hand-specified heuristics.

6 Discussion

Procedural basis versus authored error mice. The construct-validity result is strongest for the procedural basis, where a single instruction (repair an axis) produces the predicted directional movement. The calibration result is more nuanced: the basis covers truthful and underbidding behavior on its own, but overbidding—a first-order regularity in human auction data (Kagel and Levin, 1993)—is recovered only with hand-specified heuristic strains. This distinction matters for interpretation. The model-agnostic claim applies cleanly to the procedural axes; the error mice are a complementary, more bespoke device for spanning the mistake space. The two should be reported as distinct epistemic objects.

Model dependence. Because deficits are defined as procedural controls rather than artifacts of one model, the protocol can in principle be rerun on the next frontier model. The present results use a single base model; a portability check on an alternate reasoning model is the natural robustness test and is left to subsequent work.

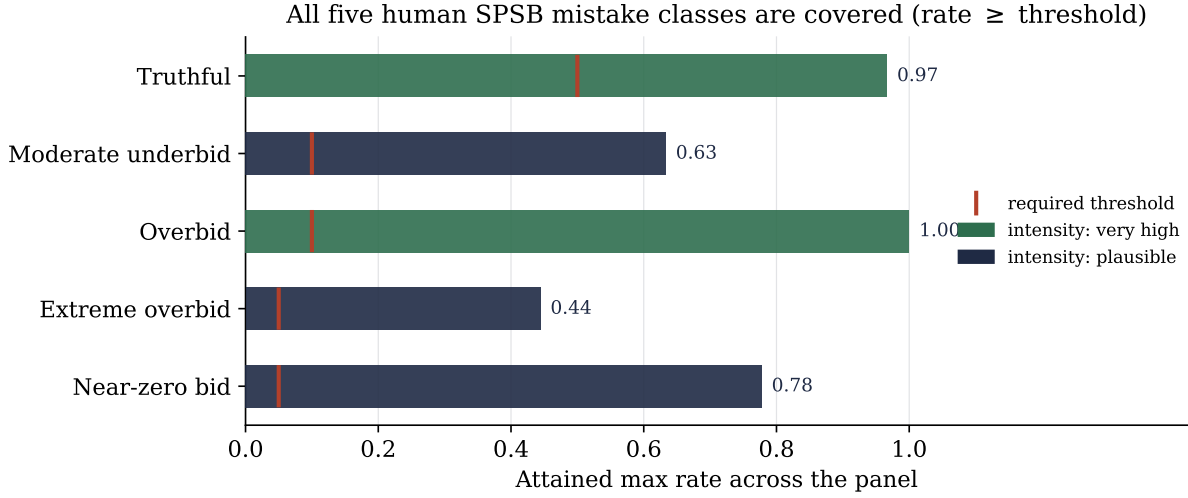


Figure 3: Calibration coverage. For each human-mistake class, the maximum rate attained by any strain in the panel (bar) against the required detection threshold (red marker). Bar color encodes the intensity band. All five classes clear their threshold.

Table 3: Calibration scorecard for the five canonical SPSB mistake classes. “Max rate” is the highest rate attained by any strain; “best strain” is the strain attaining it. All classes are covered.

Mistake class	Metric	Thresh.	Max rate	Best strain	Intensity	Covered
Truthful	truthful rate	0.50	0.967	c1_f0_b0	very high	✓
Moderate underbid	moderate underbid rate	0.10	0.633	c0_f0_b0	plausible	✓
Overbid	overbid rate	0.10	1.000	overgeneralizer [†]	very high	✓
Extreme overbid	extreme overbid rate	0.05	0.444	overgeneralizer [†]	plausible	✓
Near-zero bid	near-zero bid rate	0.05	0.778	payment panic [†]	plausible	✓

Notes. Coverage rate = 1.00 (5 of 5). [†] Supplemental error strain (spsb_error_second_price_overgeneralizer, spsb_error_payment_panic_near_zero); all other best strains are members of the clean 27-cell procedural basis.

External validity. Digital mice are a phase-I screening population. A strain reproducing a mistake class establishes that the panel can express that behavior, not that the human share of that behavior is matched. We therefore keep the stress-testing claim (this paper) separate from any calibrated prediction of human behavior, which would require human pilot data.

7 Conclusion

We show that a frontier reasoning model can be instructed into stable, separable cognitive deficits whose behavior in a second-price sealed-bid auction (i) moves in the theory-

predicted direction as each cognitive axis is repaired and (ii) spans the canonical space of human SPSB mistakes once a small set of targeted error strains is added. This supports a research program in which instructed language-model agents are used as a phase-I population to screen mechanisms for robustness to bounded rationality before human-subject experiments are run. The immediate next steps are a cross-model portability check and the extension of both the construct-validation and calibration protocols to a second domain (matching).

References

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